Transportation Research Part A 80 (2015) 184–196

Contents lists available at ScienceDirect

Transportation Research Part A

journal homepage: www.elsevier.com/locate/tra

What are the determinants in making people free riders in *proof-of-payment* transit systems? Evidence from Italy

Benedetto Barabino [a], Sara Salis [a,*], Bruno Useli [b]

[a] *Technomobility srl, viale Trieste 159/3, 09123 Cagliari, Italy*
[b] *CTM SpA, viale Trieste 159/3, 09123 Cagliari, Italy*

ARTICLE INFO

Article history:
Received 3 September 2014
Received in revised form 1 June 2015
Accepted 28 July 2015
Available online 27 August 2015

Keywords:
Fare evasion
Fare evader determinants
Proof-of-payment
Logistic regression models
Empirical evidence

ABSTRACT

Nowadays, in *proof-of-payment* transit systems, fare evasion is provoking strong interest in public transport companies (PTCs) due to the relevant economic losses, social inequity and increased levels of violence affecting personal security. Therefore, there is the need to recognize possible fare evaders. By using 2177 on-board personal interviews, gathered from an Italian PTC, and logistic regression models, we isolate determinants of possible free-rider passengers and, hence, those whom it might be advisable to target in order to capitalize on the effect of the application of countermeasures on fare evasion put forward by the local PTC. We show that males, younger than 26 years, with a low education level, unemployed and/or students and without an alternative mode of transport other than the bus are more probably fare evaders. Moreover, people who make trips of shorter than 15 min, who are systematic users and are not satisfied with the service are possible fare evaders. Finally, we found that a low level of inspection, knowledge of fines and previous ticket violations are determinants which make people more prone to evade fares. These outcomes are very useful, because, to the best of our knowledge, they represent the first empirical contribution showing the determinants which help evaluate the propensity to be a fare evader, in probabilistic terms. Moreover, they could help PTCs understand who might be a fare evader, in order to anticipate suitable countermeasures.

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1. Introduction

Fare evasion occurs when a passenger does not have a ticket or has the wrong one and represents a chronic problem both for massive and collective transport systems (e.g. Multisystems et al., 2002; Reddy et al., 2011). It is gaining relevance among public transport companies (PTCs) as the number of fare dodgers is well beyond the physical limit that PTCs find acceptable. Fare evasion mainly causes problems due to: lost fare revenues and damage to corporate image, social inequity and low levels of security. Based on data provided by 31 PTCs from 19 states worldwide, representing about 300,000 employees and 20 billion passengers per year, a study estimated that the phenomenon costs PTCs over 1 billion euros per year (Bonfanti and Wagenknecht, 2010). Moreover, fare evasion is at the root of a social inequity system, aggravating PTCs' subsidization needs and forcing paying passengers to increase their expenditure in order to counteract the financial effects of non-payment by evaders (e.g. Abrate et al., 2008). In addition, it reduces security levels because the actions employed to tackle it (e.g. increasing the number of inspectors) can trigger violence from fare evaders (e.g. Bijleveld, 2007).

* Corresponding author. Tel.: +39 070 2091285; fax: +39 070 2091222.
E-mail addresses: bbarabino@gmail.com (B. Barabino), sara.salis@technomobility.it (S. Salis), bruno.useli@ctmcagliari.it (B. Useli).

http://dx.doi.org/10.1016/j.tra.2015.07.017
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The study of fare evasion has mainly attracted scholars of security (e.g. crime) and economy, denoting a little multi-disciplinary scholarship to the argument and often presenting fascinating and different findings. From the *security* point of view, fare evasion can be included in the umbrella of studies that have analysed the application of possible countermeasures to face it. Recent studies have been based on field experiments and they have contributed to shedding additional light on the application of enforcement strategies (e.g. Bijleveld, 2007; Clarke et al., 2010) or certainty of being caught (e.g. Killias et al., 2009) and have revealed some trends in fare evasion. In a controlled experiment on trains in the Netherlands, Bijleveld (2007) showed that the standard procedure to recover the cost of the unpaid ticket is more effective than the penalty collection procedure. In a Canadian case study on light rail transit, Clarke et al. (2010) showed that changes in the level of enforcement did not illustrate clear trends in changes in the weekly evasion rates. In a Swiss case study on suburban trains, Killias et al. (2009) showed that the certainty of being caught (i.e. the probability of having one's ticket checked) worked as a deterrent in reducing fare evasion, as well as security on board. From the *economic* point of view, fare evasion can be included in the umbrella of studies that have mainly analysed how to protect PTCs from economic losses. Recent studies have often focused on: (i) establishing the optimum level of inspection (i.e. the number of inspectors); (ii) organizing the scheduling of inspection teams in terms of timing and places; and (iii) evaluating the optimal choices of fare collection systems in public transport. More precisely, some approaches have been proposed concerning (i), such as: the probability of being caught evading the fare compared to the amount of the fine (e.g. Bootheway, 2009; Kooreman, 1993), the maximization of the PTC's profit (e.g. Barabino et al., 2013a, 2014; Boyd et al., 1989) and the simulation of a situation where an inspector checks whether a passenger abides by the rules of the game theory (e.g. Avenhaus, 2004). Looking at (ii), Yin et al. (2012) focused their efforts on analysing the inspection in more rigorous ways, and formulated an optimization model to establish the organization of scheduling inspection teams. In addition, Borndörfer et al. (2012, 2013) and Correa et al. (2014) focused on models for optimizing fare inspection strategies and studied the interaction between fare inspectors and users of a transportation network. Focusing on (iii), Sasaki (2014) provided a mathematical framework to evaluate which can be more profitable for a PTC: barrier or barrier-free fare collection systems.

Generally speaking, fare evasion can be classified according to the broad research on the unethical behaviour of people (e.g. Gneezy, 2005; Mazar et al., 2008; Suquet, 2010), as it somewhat represents its kind of specification. Very few studies have analysed personal determinants in order to derive the fare-evader "profile" (i.e. detecting determinants of who is more likely to evade the fare). Using an unobtrusive observation methodology, Eddy (2010) collected data on gender and demographic age of 280 people evading fares at Melbourne metropolitan stations, and reported, by using descriptive statistics, that people committing fare evasion are largely male and adult. By a direct measure of the ticket-holder (i.e. tickets and passes, of course), Abrate et al. (2008) and Bucciol et al. (2013) showed some relevant determinants of fare evaders, by using logistic regression models (LRMs). Based on 16,000 observations related to a sample of 12 Italian PTCs, Abrate et al. (2008) isolated a few socio-demographic characteristics of fare evaders. Using the logit model, they showed that the younger, unemployed and foreigners (i.e. not Italian) are the more probable fare evaders. A more refined work is presented in Bucciol et al. (2013). Indeed, they detected the individual determinants that predict unethical behaviour, supporting their argumentations on the topic of fare evasion. They collected data by using a two-step methodology focusing on: (i) basic socio-demographic information gathered by a visual estimate on 541 passengers and, (ii) specific behaviour of the person committing fare evasion gathered by a direct survey of 144 passengers. Using the probit model, they showed that young, male and non-European immigrants in the sample (citizens of Reggio Emilia) are more likely to be fare evaders. Yet, Mehlkop et al. (2007), in the context of the "Theory of Crime", model the utility function related to attitude to tax and fare evasion and link them to categorical determinants. Using a stated preference survey, respondents were asked hypothetical questions on: benefits resulting from evasion, the probability of being an evader, the severity of fines, the probability of being detected, the attitude towards norms and risk-taking. Based on more than 1900 interviews in Dresden and Berne, they showed that the attitude towards evasion is a function of the subject's knowledge of the phenomenon, while neglecting his/her socio-demographic characteristics.

Although all these studies added contributions, two relevant considerations arise. First, fare evasion is often faced focusing on the side of the PTC rather than that of the fare-evader. Indeed, security and economic studies opened doors to analyse the fare evasion issue on the principle of how to face it and/or protect the PTC from economic losses. However, these studies did not recognize who are the fare evaders. Second, no previous research has returned empirical data, in a single study, on the type of people who will probably commit fare evasion and their socio-demographic group, travel behaviour characteristics and personal knowledge of fare evasion. This integrated kind of data is somewhat ignored and missed from the research findings. Indeed, to the best of our knowledge, even if the work of Bucciol et al. (2013) can be considered the more refined study on this topic, it uses few data to evaluate the evader "profile". More important, they begin the analysis starting from caught-evader data and do not evaluate a possible propensity to evade the fare, by a direct question in a hypothetical scenario, even if they recognize the possible interest. On the other hand, Abrate et al. (2008) also began the analysis starting from the caught-evader data and limited their analysis to the socio-demographic characteristics and the purpose of the trip. Besides, they express the need to investigate the link between fare evaders and the firm-specific and enforcement variables, such as the level of inspection and fines. In addition, to the best of our knowledge, most advanced PTCs collect data on fare evaders, but they derive the fare-evader "profile", if any, by manipulating the report on fines, which contains few data about gender, age, and time of detection, while presenting outcomes by descriptive statistics.

Therefore, it is crucial to shed new light on the possible fare-evader "profile", making use of models and statistical data, thus answering the following research question:

What are the determinants in making people fare evaders in *proof-of-payment* transit systems?

The contribution of this paper is to answer this question using LRMs to evaluate the propensity for the passenger to evade the fare. The LRM is based on field-collected data through on-board surveys on a large sample of people travelling across all bus routes of an Italian PTC network. Answering the research question is relevant for PTCs, as they are put in a favoured position to know who are the possible fare evaders and, hence, to anticipate suitable countermeasures. For example, PTCs can minimize their economic efforts of inspection (thus saving valuable resources, which can be used elsewhere) by checking people who are mainly expected to be fare evaders. Moreover, there is interest in subsidizing agencies in order possibly to revise the norms related to the fare and the amount of the fine, which can affect the fare-evader "profile". For example, subsidizing agencies could re-think new fares and/or fines to account for the people committing the offence. In addition, as appointed by Bijleveld (2007), detecting relevant determinants of evaders is recommended due to the common problem of unpaid fines, to better differentiate people who never pay the fine from those who might pay, thus contributing to revenues. Finally, according to Bucciol et al. (2013) measuring the marginal effects of some determinants can help in understanding whether the statistically significant determinants affect the costs of the PTC as well.

The remainder of this paper is organized as follows. In Section 2, materials are presented and methods applied to a bus network managed by a medium-sized PTC, in order to derive the fare-evader "profile". In Section 3, results are discussed and largely compared with previous works. Finally, in Section 4, conclusions and research perspectives are provided.

2. Materials and methods

To assist in answering the research question, a set of criteria was considered: what data are required, how and where will they be collected and, then analysed.

2.1. Data type and collection method

Data for the study were gathered through an on-board survey among passengers of the Cagliari bus system. Encompassing eight communalities for a total of approximately 350,000 people, the area of study is characterized by the highest demographic density in Sardinia. In addition, it represents the island's main commercial and administrative hub, attracting dozens of thousands of commuters on a daily basis. Urban public transportation within the area is guaranteed by the local PTC, namely CTM, on 29 urban routes.

The main survey's purpose is the evaluation of the propensity to be a fare evader. Theoretically, the propensity to be a fare evader may depend on many determinants, including socio-demographic characteristics, features relating to the trip and to the public transport system chosen, as well as personal knowledge of fare evasion. Therefore, we built a four-section questionnaire and gathered data on various aspects.

- The first section deals with socio-demographic characteristics such as: gender, age, educational qualification, employment, car availability, the holding of a driving licence and reasons for using the bus.
- The second section deals with the travel behaviour characteristics, such as: origin, purpose of the trip, time of day, in-vehicle time, use of other transit systems, travel frequency and overall rate of satisfaction with the service.
- The third section deals with personal knowledge of fare evasion, such as: perception of the inspection level in a predefined time window (revealed inspection frequency), knowledge of the amount of the fine and previous ticket violations (i.e. whether the passenger had already been fined).
- The fourth section deals with the question on the propensity to be a fare evader, which is evaluated in a hypothetical scenario namely "without inspection" and acts as a response variable. It was as follows: if no checks are performed, would you buy a ticket? In our opinion, this question seemed to be appropriate to capture that segment of users showing an innate inclination to be fare evaders, irrespective of the actual choice to evade the fare or not. In fact, it is possible to perceive a trend in the dishonest behaviour of passengers towards compliance with travel rules, regardless of their actual behaviour at that moment. One may be inclined to consider that the estimation of the propensity to be a fare evader using this question may be biased on the high side, because it seems plausible that some customers would lie in response to the direct question. However, we will refer to it in the model, even if a more refined estimation could be investigated in the future. Moreover, one could argue that respondents may be reluctant to acknowledge being potential fare evaders without personal gains. Indeed, most of the economic theory assumes that people misreport their private information if this is to their material benefit (e.g. Maskin, 1977; Matsushima, 2008). Conversely, recent studies rely on the assumption that people can experience a psychological disutility, which holds them back from misreporting, at least to some extent. They rise up many reasons of do not misreporting such as intrinsic lying costs, honesty, and conditional cooperation (e.g. Abeler et al., 2014; Erard and Feinstein, 1994; Traxler, 2010).

The questionnaire framework highlighted the need to use a paper and pencil interview (PAPI) instead of a self-administered interview, in order to enhance, as much as possible, the quality, accuracy and reliability of the data collected. Hence, a crucial phase of the survey preparation stage was related to the interviewers' training activities, devised to ensure the highest possible homogeneity during the survey administration phase. In addition, in order to stimulate the

participation to the survey, we trained interviewers to be very persuasive towards respondents. In this way, the interviewers tried to induce respondents to agree in survey, through appeals to intentions, attitudes, values and actions sponsored by them.

Based on a headway route sample plan, the questionnaires were administered by 16 trained interviewers, during two weeks in March 2013 and two more weeks in July 2013, from 0700 AM to 0700 PM. Riders were first randomly approached, and then the questions were administered. Differently from Abrate et al. (2008), Bucciol et al. (2013) and Eddy (2010), who collected data at some bus stops and stations (i.e. point observation), we collected data on-board (i.e. ride observation). The motivating reason was the possibility of removing shortcomings derived from choosing check points. Moreover, observations collected at points are not representative of all possible fare evaders met along the route. Therefore, by using a ride observation, one is expected to obtain more accurate information, because one can intercept passengers along the entire route.

In addition, the on-board methodology has been preferred over other surveying techniques (mail, telephone, web, and so on) because of some advantages: the possibility to cover all of the urban routes of interest, the higher response rates, the ability to collect the information from the surveyed sample experiencing the service (TCRP Synthesis 63, 2005). The literature on "behavioural economics" disclosed that several studies use incentives to increase response rates in postal, telephone and face-to-face (at home) surveys (e.g. Simmons and Wilmot, 2004; Singer and Ye, 2013). However, this survey was performed without the use of incentives and a response rate of 75% was obtained. Nevertheless, if the response rate is high, the effect of incentive is low (Singer et al., 1999).

In our opinion, there are several possible reasons for not using incentives in this survey. First, people still answer survey questions without incentives for altruism, survey related reasons and personal reasons (Porst and Von Briel, 1995). Second, as many people can be easily approached by the interviewer, the on-board survey in public transportation helps gaining a higher response rate in comparison to other techniques (e.g. Barabino et al., 2013b). Moreover, similarly to what happens in air transport and differently from telephone or face-to-face (at home) surveys, the passenger may even welcome the survey as a constructive way to use his own time while on board (Denstadli, 2000; Richardson et al., 1995). Third, incentives seems unsuitable when they are used to induce participation on surveys in presence of avoidable or unreasonable risks (Singer and Bossarte, 2006), as it would seem to be our case. Indeed, since the survey is anonymous, performed by interviewers who do not appear as ticket inspectors, respondents are sure to avoid any risk of penalty or punishment, even when declaring their potential propensity to evade the fare. Four, the use of incentives (both monetary and not monetary) in an on-board survey can be quite expensive, especially in case of a large sample. Five, in a recent review, Singer and Ye (2013) have reported that relatively few studies evaluated the effect of incentives on the quality of responses and did not find evidence of a direct relation between incentives and surveys reliability and validity.

At the end of survey, the data collection activity produced 2177 usable questionnaires.

More precisely, Table 1 describes all the variables included in the dataset and used in the model, both response and explanatory.

The response variable is represented by the propensity to evade the fare which is denoted by *EV_SP* and is determined by the aforementioned question asked of CTM's passengers regarding their stated propensity to evade the fare for the trip, in the hypothetical context namely *If no checks are performed* (i.e. without ticket inspection activities). As a result, since in this study we can consider just two options, we consider a possible fare evader as one who would not buy the ticket, thus $EV_SP = 1$; on the other hand, a passenger not considered a possible fare evader is one who would buy a ticket anyway, thus $EV_SP = 0$.

The explanatory variables are distinguished in the following three groups based on determinants related to:

- Socio-demographic characteristics.
- Travel behaviour characteristics.
- Personal knowledge of fare evasion.

Note that two variables are continuous (bus use frequency – *Month_freq* – and subjective probability of inspection – p_i), some others are categorical (age, educational qualification, perceived inspection frequency) and the remaining dichotomous. The dummy variables – in italics in Table 1 – will not be included in the modelling process, because they are used as the reference case, in order to compare model outcomes. Finally, Table 1 also provides descriptive statistics on the sample for each dichotomous variable considered. All descriptive statistics are self explanatory.

2.2. Data analysis

The data analysis followed a two-step approach:

- Correlation analysis.
- Inferential analysis (Preliminary logistic regression analysis, stepwise logistic regression, further refinement of the model).

Table 1
Response variable and explanatory variables used in the model.

Response variable	Abbreviation	Description	%
Evasion propensity	EV_SP	Stated probability of being a fare evader	28.20[c]
	Dummy variable	*Stated probability of not being a fare evader*	*71.80*
Explanatory variables			
Socio-demographic characteristics			
Gender	Gen_F	Female	61.78
	Dummy variable	*Male*	*38.22*
Age	Above_50	Above 50 years old	15.89
	26_50	Between 26 and 50 years old	34.09
	Dummy variable	*Under 26 years old*	*50.02*
Educational qualification	Upper_sc	Upper school graduated	58.47
	Middle_sc	Middle school graduated	38.18
	Dummy variable	*Middle school not graduated*	*3.35*
Employment	Work_y	Worker	29.54
	Dummy variable	*Not worker*	*70.46*
	Stud_y	Student	42.81
	Dummy variable	*Not student*	*57.19*
	Unempl_y	Unempolyed	13.18
	Dummy variable	*Not unemployed*	*86.82*
Holds a driving licence	Drive_lic_y	Holds a driving licence	55.81
	Dummy variable	*Does not hold a driving licence*	*44.19*
Car availability	Car_y	Has a car	32.02
	Dummy variable	*Does not have a car*	*67.98*
Reason to use bus	Other_use_bus	Use of the bus for other reasons (not related to the lack of trip alternatives)	37.80
	Dummy variable	*Use of the bus because there are no alternatives*	*62.20*
Travel behaviour characteristics			
Trip purpose	Syst_trips	Systematic trips for work or study	47.59
	Dummy variable	*Not systematic trips (for other purposes)*	*52.41*
	Leis_trips	Leisure trips (shopping, sport, amusement, etc.)	19.11
	Dummy variable	*Not leisure trips*	*80.89*
Time of the day	Rush_hour	Rush hours trips (7.00–9.00 and 13.00–14.00)	20.49
	Dummy variable	*Not rush hours trips*	*79.51*
Trip origin	Orig_CA	Trip origin is in Cagliari	45.57
	Dummy variable	*Trip origin is not in Cagliari*	*54.43*
In-vehicle time	In_vehicle_time_more15	Travel time more than 15 min	73.45
	Dummy variable	*Travel time less than 15 min*	*26.55*
Other transit systems use	Other_transit_y	Use of other transit systems	38.54
	Dummy variable	*No use of other transit systems*	*61.46*
Bus use frequency	Month_freq[a]	Average monthly number of trips	–
	Daily_trips_over2	Number of daily trips over 2	38.63
	Dummy variable	*Number of daily trips under 2*	*61.37*
Quality rating	Satisf_y	Satisfied user (grade on the overall service more than sufficient)	94.95
	Dummy variable	*Not satisfied user (vote on the overall service more than sufficient)*	*5.05*
Personal knowledge of fare evasion			
Perceived inspection frequency	Insp_freq_more5	The user has seen the inspectors boarding the vehicle to perform control activities more than 5 times in 4 months	17.18
	Insp_freq_1_5	The user has seen the inspectors boarding the vehicle to perform control activities a number of times between 1 and 5 in 4 months	51.81
	Dummy variable	*The user has never seen the inspectors boarding the vehicle to perform control activities in 4 months*	*31.01*
Subjective probability of inspection	p_i[b]	Ratio between perceived inspection frequency and travel frequency	–
Amount of the fine	Know_fine_y	The user knows the amount of the fine	62.66
	Dummy variable	*The user does not know the amount of the fine*	*37.34*
Fines in the past	Fine_past_y	The user has already been fined in the past	42.81
	Dummy variable	*The user has not been fined in the past*	*57.19*

[a] Continuous variable. Mean: 41.9345 Std. Dev.: 32.2213.
[b] Continuous variable. Mean: 0.0785 Std. Dev.: 0.1257.
[c] In this study, we found statistical significance (p-value < 0.001) on the fact that respondents hardly lie, even without incentives. Indeed, we also asked respondents whether they had a valid ticket, whilst not controlling for it. The 12.78% of the passengers were evaders, while, when asking the question in the hypothetical context (i.e. without ticket controls) the percentage increased up to 28.20%. This is an expected result, since it is plausible that the number of potential evaders is higher than the one actually detected.

The first step reports the result of a correlation analysis in order to detect the mutual statistical dependence between couples of the determinants investigated. In this way, one is able make a preliminarily check as to whether multi-collinearity problems occur.

The second step provides the outcomes of the main determinants, which possibly proceed to the building of the fare-evader "profile" subject, by using statistical models. As shown in Section 2.1, *EV_SP* is binary. In these cases, statistical models such as the generalized linear, probit and logit help explain the possible choice of fare evasion, in a probabilistic way. Models for binary data ordinarily assume a binomial distribution for the response variable. Logit can be considered a canonical model appropriate to describe the propensity to evade. Indeed, even when compared to other generalized linear models, it has more intuitive interpretation of the results in terms of odds ratio (OR), which are a measure of association between the probability of a success (the propensity to evade the fare) and the one of a failure (not to evade the fare). In other words, they measure the number of successes for each failure.

More formally, the LRM can be derived by a sigmoid curve, whose equation is:

$$P(y_i = 1|x_i) = \pi_i = \frac{e^{\beta_0+\beta_1 x_1+\ldots+\beta_n x_n}}{1+e^{\beta_0+\beta_1 x_1+\ldots+\beta_n x_n}} \tag{1}$$

where

- y_i is the response variable,
- π_i represents the probability of a success,
- x_i are the explanatory variables which help evaluate the propensity of being a fare evader,
- $\beta_0 \ldots \beta_n$ represent the model unknown coefficients and indicate whether the logistic curve goes up or down, as x_i increases.

In the considered case, Eq. (1) returns the probability of being a potential fare evader, for the *i*th user.

By manipulating the logistic function one observes that the logarithm of the odds – namely *logit* (π_i) – is the link function between the response variable y_i and the linear combination of explanatory variables x_i, to ensure that the response y_i will always be in the range [0–1], for each value assumed by the explanatory variables. As a result:

$$\text{logit}(\pi_i) = \log\left[\frac{\pi_i}{1-\pi_i}\right] = \log(\text{odds}) = (\beta_0 + \beta_1 x_1 + \ldots + \beta_n x_n) \tag{2}$$

The estimation of the model is based on a maximum likelihood procedure using the software Genstat®. The evaluation of the model will be based on using the deviance ratio (*d.r.*), which represents the ratio between the regression deviance and the degree of freedom of the model, and by checking its statistical significance. Moreover, the signs of the coefficients and their significance are also evaluated.

In the following section, the outcomes of these analyses are presented.

3. Results and discussion

3.1. Correlation analysis

In order to detect whether there are possible correlations between determinants, we performed a correlation analysis among all explanatory variables. The results are shown in Table 2, which reports values and signs of correlation indexes between a pair-comparison of all determinants. Bold values represent correlation indexes greater than 0.5, which, to our knowledge, can be considered as a threshold separating strong from possible weak correlations. In Table 2, the 2-tailed significances are given in parenthesis and they are in bold type at the 0.001 level. The correlation matrix presented in Table 2 shows that multi-collinearity potential problems appear not to exist. Moreover, it is worth noting that most of the correlation indexes have rather small values. However, for some of the 21 pairs of variables, the pair-wise correlation indexes have values greater than 0.5 which are also significant at the 0.001 level and deserve some consideration. First, there is a strong and positive correlation (0.880) between the two dichotomous variables *Car_y* and *Other_bus_use*. This is a somewhat expected result: in fact, car owners probably do not travel by bus because of lack of alternatives, but for other reasons. Second, even if less pronounced, Table 2 shows other positive correlations, between *Car_y* and *Drive_lic_y* (0.611) and between *Other_bus_use* and *Drive_lic_y* (0.604). In this case, the outcomes can be explained because those who have a driving licence probably have a car and those with a driving licence probably travel by bus not due to lack of alternatives, but for other reasons (e.g. unavailability of car parking at the destination). Third, two negative correlations were observed between *Stud_y* and Age (−0.650) and between *Stud_y* and *Work_y* (−0.560). While the first correlation probably expresses a link between employment status and the specific age group, the second indicates that it is unlikely that students are also workers. Based on the correlation outcomes, among *Car_y*, *Drive_lic_y* and *Other_bus_use* determinants, it is likely that only one will be considered in the model.

However, correlation indexes do not necessarily disallow the aforementioned variables from appearing together in the model: they just highlight if there are multi-collinearity problems. In this way, the correlation moves towards the possible reduction of the number of possible explanatory variables in the model, according to the principle of parsimony. Nevertheless, previous analyses do not take into account the specific characteristics of passengers who will probably commit

Table 2
Correlation matrix.

	Gen_F	Age[a]	Educat. qualif.[a]	Unempl_y	Work_y	Stud_y	Drive_lic_y	Car_y	Other_use_bus	Syst_trips	Leis_trips	Rush_hour	Orig_CA	In_vehicle_time_more15	Other_transit_y	Month_freq[b]	Satisf_y	Perc.insp. freq.[a]	p_i[b]	Know_fine_y	Fine_past_y
Gen_F	1.000	0.135	0.082	−0.076	0.014	−0.072	0.025	−0.048	−0.040	0.015	−0.029	0.048	−0.015	0.032	−0.005	−0.087	−0.009	0.006	0.033	−0.083	−0.107
	(<0.001)	**(<0.001)**	**(<0.001)**	**(<0.001)**	(0.5149)	**(<0.001)**	(0.2361)	(0.0255)	(0.0627)	(0.4836)	(0.1779)	(0.0254)	(0.4855)	(0.1316)	(0.8312)	**(<0.001)**	(0.6815)	(0.7972)	(0.1192)	**(<0.001)**	**(<0.001)**
Age[a]		1.000	−0.003	−0.042	0.289	**−0.650**	0.296	0.323	0.335	−0.153	−0.100	0.000	0.113	−0.017	−0.167	−0.232	0.041	−0.087	0.144	−0.350	−0.263
		(<0.001)	(0.8991)	(0.0475)	**(<0.001)**	**(<0.001)**	**(<0.001)**	**(<0.001)**	**(<0.001)**	**(<0.001)**	**(<0.001)**	(0.9875)	**(<0.001)**	(0.4192)	**(<0.001)**	**(<0.001)**	(0.0548)	**(<0.001)**	**(<0.001)**	**(<0.001)**	**(<0.001)**
Educat. qualif.[a]			1.000	0.019	0.204	−0.034	0.446	0.281	0.293	0.241	−0.186	−0.010	0.005	−0.004	0.063	−0.158	−0.016	−0.048	0.071	−0.120	−0.135
			(<0.001)	(0.3784)	**(<0.001)**	(0.1093)	**(<0.001)**	**(<0.001)**	**(<0.001)**	**(<0.001)**	**(<0.001)**	(0.6467)	(0.8066)	(0.8358)	(0.0032)	**(<0.001)**	(0.4464)	(0.0243)	**(<0.001)**	**(<0.001)**	**(<0.001)**
Unempl_y				1.000	−0.252	−0.337	0.013	−0.087	−0.071	−0.322	0.014	−0.026	0.003	0.050	−0.018	0.012	−0.040	0.011	0.017	0.071	0.110
				(<0.001)	**(<0.001)**	**(<0.001)**	(0.5386)	**(<0.001)**	**(<0.001)**	**(<0.001)**	(0.5031)	(0.2212)	(0.8766)	(0.0201)	(0.3900)	(0.5724)	(0.0602)	(0.5669)	(0.4168)	**(<0.001)**	**(<0.001)**
Work_y					1.000	**−0.560**	0.270	0.341	0.345	0.308	−0.181	0.023	0.004	0.020	−0.020	−0.078	0.011	0.010	0.075	−0.179	−0.053
					(<0.001)	**(<0.001)**	**(<0.001)**	**(<0.001)**	**(<0.001)**	**(<0.001)**	**(<0.001)**	(0.2808)	(0.8503)	(0.3539)	(0.3440)	**(<0.001)**	(0.5935)	(0.6337)	**(<0.001)**	**(<0.001)**	(0.0126)
Stud_y						1.000	−0.277	−0.303	−0.324	0.181	0.125	0.014	−0.076	−0.018	0.154	0.172	0.009	0.033	−0.147	0.276	0.111
						(<0.001)	**(<0.001)**	**(<0.001)**	**(<0.001)**	**(<0.001)**	**(<0.001)**	(0.5155)	**(<0.001)**	(0.4018)	**(<0.001)**	**(<0.001)**	(0.6792)	(0.1288)	**(<0.001)**	**(<0.001)**	**(<0.001)**
Drive_lic_y							1.000	**0.611**	**0.604**	0.127	−0.205	−0.018	−0.003	0.012	−0.052	−0.268	0.019	−0.092	0.199	−0.264	−0.221
							(<0.001)	**(<0.001)**	**(<0.001)**	**(<0.001)**	**(<0.001)**	(0.3975)	(0.8869)	(0.5853)	(0.0157)	**(<0.001)**	(0.3871)	**(<0.001)**	**(<0.001)**	**(<0.001)**	**(<0.001)**
Car_y								1.000	**0.880**	0.050	−0.136	−0.017	0.025	0.007	−0.129	−0.262	0.059	−0.109	0.199	−0.309	−0.224
								(<0.001)	**(<0.001)**	(0.0199)	**(<0.001)**	(0.4396)	(0.2531)	(0.7507)	**(<0.001)**	**(<0.001)**	(0.0056)	**(<0.001)**	**(<0.001)**	**(<0.001)**	**(<0.001)**
Other_use_bus									1.000	0.050	−0.133	−0.006	0.025	0.008	−0.109	−0.243	0.041	−0.098	0.201	−0.291	−0.211
									(<0.001)	(0.0197)	**(<0.001)**	(0.7754)	(0.2495)	(0.7256)	**(<0.001)**	**(<0.001)**	(0.0531)	**(<0.001)**	**(<0.001)**	**(<0.001)**	**(<0.001)**
Syst_trips										1.000	0.027	0.109	−0.100	−0.019	0.094	0.059	0.018	0.087	−0.114	0.038	0.006
										(<0.001)	**(<0.001)**	**(<0.001)**	**(<0.001)**	(0.3853)	**(<0.001)**	(0.0055)	(0.3946)	**(<0.001)**	**(<0.001)**	(0.0776)	(0.7632)
Leis_trips											1.000	−0.099	0.057	0.017	−0.027	0.027	0.027	−0.028	−0.010	0.054	0.040
											(<0.001)	**(<0.001)**	(0.0075)	(0.4262)	(0.2049)	(0.2046)	(0.2117)	(0.1867)	(0.6406)	(0.0117)	(0.0626)
Rush_hour												1.000	−0.012	−0.032	0.033	0.014	−0.018	−0.026	−0.020	0.039	0.012
												(<0.001)	(0.5773)	(0.1303)	(0.1237)	(0.5046)	(0.4012)	(0.2315)	(0.3458)	(0.0692)	(0.5872)
Orig_CA													1.000	−0.162	−0.205	0.051	−0.016	0.043	−0.069	−0.054	0.010
													(<0.001)	**(<0.001)**	**(<0.001)**	(0.0179)	(0.4466)	(0.0440)	(0.0013)	(0.0111)	(0.6442)
In_vehicle_time_more15														1.000	−0.015	0.024	−0.020	0.057	0.002	0.063	0.041
														(<0.001)	(0.4703)	(0.2711)	(0.3517)	(0.0081)	(0.9308)	(0.0034)	(0.0565)
Other_transit_y															1.000	0.063	−0.007	0.082	−0.025	0.092	0.089
															(<0.001)	(0.0030)	(0.7468)	**(<0.001)**	(0.2466)	**(<0.001)**	**(<0.001)**
Month_freq[b]																1.000	−0.045	0.262	−0.443	0.265	0.251
																(<0.001)	(0.0367)	**(<0.001)**	**(<0.001)**	**(<0.001)**	**(<0.001)**
Satisf_y																	1.000	−0.047	0.006	−0.031	−0.050
																	(<0.001)	(0.0287)	(0.772)	(0.1523)	(0.0185)
Perc.insp.freq.[a]																		1.000	0.012	0.157	0.143
																		(<0.001)	(0.5842)	**(<0.001)**	**(<0.001)**
p_i[b]																			1.000	−0.190	−0.149
																			(<0.001)	**(<0.001)**	**(<0.001)**
Know_fine_y																				1.000	0.409
																				(<0.001)	**(<0.001)**
Fine_past_y																					1.000
																					(<0.001)

[a] Categorical variables.
[b] Continuous variables.

the offence. Thus, in the remainder of the paper we will shed new light on it, considering all these determinants jointly and, starting from the full model (i.e. the model with all explanatory variables), we will undertake further refinements, to explain, with major insights, the statistical inferences.

3.2. Inferential analysis

When descriptive analysis is undertaken, we concentrate our efforts on the detection of the main determinants of people committing fare evasion. At first, some changes to the databank structure were made, in order to return in most cases categorical variables. Then, the dependent and independent variables were introduced to run the model. The outcomes of the LRM are presented in Table 3.

Three models are presented: the full, the reduced after a substantial improvement and the final after a further refinement. Moreover, for each model the coefficients (Estimate), their significance (*p*-value), in bold type when it is <.001, and the odds ratio (OR) are represented. Finally, the last part of Table 3 reports the summary statistics.

Generally speaking, all models fitted well to the data. Indeed a large *d.r.* coupled with a small *p*-value indicates a "good" goodness-of-fit of the logistic model: the χ^2 is consistently <.001 showing a good fit of the data for each model. The models will be discussed in the following three sections.

Table 3
Model coefficient estimates and summary statistics.

Determinant	Full Model			Reduced Model			Refined Model		
	Estimate	*p*-value	OR	Estimate	*p*-value	OR	Estimate	*p*-value	OR
Constant	0.438	0.433	1.549	0.894	0.066	2.444	1.029	0.032	2.798
Gen_F	−0.363	0.002	0.6953	−0.379	**0.001**	0.6846	−0.392	**<.001**	0.6758
Age									
Above_50	−1.950	**<.001**	0.1422	−2.151	**<.001**	0.1164	−2.183	**<.001**	0.1127
26_50	−1.056	**<.001**	0.3478	−1.084	**<.001**	0.3382	−1.111	**<.001**	0.3292
Educational qualification									
Upper_sc	−1.411	**<.001**	0.2439	−1.305	**<.001**	0.2711	−1.338	**<.001**	0.2623
Middle_sc	−1.008	0.009	0.3649	−0.891	0.016	0.4103	−0.901	0.014	0.4061
Employment									
Work_y	0.513	0.140	1.671						
Stud_y	0.795	0.027	2.215	0.358	0.045	1.430	0.347	0.051	1.415
Unempl_y	0.824	0.020	2.279	0.416	0.033	1.516	0.395	0.042	1.485
Drive_lic_y	−0.106	0.503	0.8992						
Car_y	−0.189	0.519	0.8281						
Other_use_bus	−0.259	0.324	0.7719	−0.456	0.002	0.6341	−0.476	0.001	0.6211
Trip purpose									
Syst_trips	−0.086	0.599	0.9180						
Leis_trips	−0.032	0.857	0.9689						
Rush_hour	0.185	0.195	1.203						
Orig_CA	0.212	0.080	1.236						
In-vehicle time_more15	−0.301	0.023	0.7404	−0.346	0.008	0.7073	−0.347	0.008	0.7065
Other_transit_y	0.194	0.108	1.214						
Bus use frequency									
Month_freq	0.00382	0.032	1.004	0.00438	0.006	1.004			
Daily_trips_over2							0.224	0.055	1.251
Satisf_y	−1.190	**<.001**	0.3044	−1.196	**<.001**	0.3024	−1.211	**<.001**	0.2979
Perceived inspection frequency									
Insp_freq_more5	−0.490	0.010	0.6129	−0.493	0.008	0.6107	−0.444	0.016	0.6413
Insp_freq_1_5	−0.259	0.050	0.7715	−0.264	0.044	0.7680	−0.241	0.065	0.7857
p_i	−0.132	0.833	0.8763						
Know_fine_y	0.971	**<.001**	2.641	0.978	**<.001**	2.660	1.049	**<.001**	2.856
Fine_past_y	1.002	**<.001**	2.723	1.030	**<.001**	2.801	0.992	**<.001**	2.698

Summary Statistics	Degree of freedom	Deviance	Mean deviance	Degree of freedom	Deviance	Mean deviance	Degree of freedom	Deviance	Mean deviance
Regression	24	728	30.3211	15	717	47.8288	15	713	47.5608
Residual	2152	1862	0.8654	2161	1873	0.8666	2161	1877	0.8684
Total	2176	2590	1.1903	2176	2590	1.1903	2176	2590	1.1903
d.r	*30.32*			*47.83*			*47.56*		
χ2	***<.001***			***<.001***			***<.001***		

3.2.1. The full model

Generally speaking, the full model properly fits the data. Indeed, the statistical test on *d.r.* produces a small *p*-value for goodness-of-fit (<0.001). Therefore, there seems to be strong evidence of a regression effect here (i.e. not all the β_j are zero). It is worth noting that some of the variables are very significant and some are not. Looking at the socio-demographic characteristics, the variables gender, age and education greatly help explain the propensity to evade fares. On the other hand, the high *p*-value on the *t*-test performed shows a probable lack of capacity for an explanation of variables *Car_y* and *Drive_lic_y*. The level of employment and the use of the bus for other reasons can help explain the fare evasion propensity a little. Focusing on the travel behaviour characteristics, some unexpected trends can be observed. In fact, the in-vehicle time and satisfaction with the service appear partially useful in explaining the propensity to evade fares. On the other hand, the reason for the trip, the time of the trip and the place of origin appear unable to explain this phenomenon in our data set. Therefore, even if the fare evader pays attention to the in-vehicle time and to the satisfaction with the service, he/she does not belong to a specific class regarding the reason for the trip, the time and place of origin: he/she can be detected regardless. Finally, considering the personal knowledge of fare evasion, knowledge of the amount of the fine and previous ticket violations (i.e. having already been fined) seem to greatly affect the passenger in his choice to evade the fare, as well as the inspection frequency, even if to a lesser extent. A brief examination of the signs of the coefficient shows some expected results. Effectively, not being too young, having an upper school qualification and being satisfied with the transit system appear features that reduce the probability of being a fare evader, as well as the increase in inspection frequency; conversely, knowing the amount of the fine and having already been fined increase the propensity to evade the fare.

3.2.2. The stepwise procedure and the reduced model

According to the principle of parsimony and in order to build a more improved model, controlling the variables entering or exiting within the LRM, Genstat provides different procedures. By them, it is possible to eliminate all unnecessary variables from the regression, in order to achieve a balance between reduced complexity and an adequate goodness-of-fit of the model. In this specific study, stepwise regression analysis is used. The stepwise procedure enables the current regression model to be changed by adding or removing explanatory variables, in order to achieve the greatest improvement. The procedure ends when the allowed maximum number of steps is reached. The maximum number of steps is a Genstat input and it is suggested that it should be equal to 2 * degree of freedom (*d.f.*) of the regression, as every variable is added or dropped according to the improvement obtained by the model based on *d.r.* After the application of the stepwise procedure, according to the socio-demographic characteristics, travel behaviour characteristics and personal knowledge of fare evasion, the variables dropped (i.e. those held not to be significant in explaining the propensity to evade) were: *Work_y*, *Car_y*, *Drive_lic_y*, *Leis_trips*, *Syst_trips*, *Rush_hour*, *Orig_CA*, *Other_transit_y* and p_i. Looking at these outcomes one focuses on various points.

First, the stepwise regression helped remove variables with a strong correlation (i.e. *Work_y*, *Car_y*, *Drive_lic_y*), so that they were not included in the final model.

Second, the trip purpose -*Leis_trips* and *Syst_trips*-, the trip time -*Rush hour*-, the use of other public transport systems -*Other_transit_y*- and the origin of the trip -*Orig_CA*- do not seem to greatly affect the probability of being an evader. At first glance, the exclusion of the trip purpose and the trip time appears barely comprehensible. Maybe, this exclusion could be explained because to some extent the model indirectly includes other determinants which take into account why and when a user travels. For example, variables *Stud_y* and *Unempl_y* are present and can indicate the condition of being a student and unemployed, respectively. Indeed, it is likely that students travel mostly during peak hours and for study purposes. On the contrary, the unemployed people travel at unspecified times – often off-peak hours – and for reasons different from study or work. These outcomes are consistent with previous studies. Bucciol et al. (2013) showed that the trip purpose expressed as "travel for pleasure" has no significant effect on the probability of evasion. Then, the elimination from the final model can be justified. In addition, Abrate et al. (2008) and Bucciol et al. (2013) found that the time period in which the trip occurs is not statistically significant in explaining the effect of being a fare evader.

Third, the use of other transit systems, as well as the origin of the passenger – that is, if he starts his trip from the main city (Cagliari) or from somewhere else in the metropolitan area – does not seem to significantly affect the variability of the propensity for evasion, probably due the lack of a detailed segmentation of the traffic zone. Then, the variable is removed from the final model. Finally, the variable p_i is obtained as the ratio between perceived inspection frequency and travel frequency, which are already included in the model with perceived inspection frequency and *Month_freq*. Therefore, dropping p_i after the stepwise procedure is a somewhat expected outcome, likely due to the redundancy of information of the same type in explaining the phenomenon.

The results of the reduced model aim to explain some trends in the fare-evader profile. As with the full model, the reduced model produces a goodness-of-fit. Indeed, a high *d.r.* and a small *p*-value are detected. However, the new model is much less complex (d.f. reduced by 9 units = 24 − 15) and presents a *d.r.* higher than that of the previous one (47.83 *vs* 30.32). Moreover, all explanatory variables have coefficients with a *t-test* value greater than 2.00, so the *p-value* is low, resulting in a high capacity to explain the propensity of being a fare evader. The signs of the coefficients are all as expected. Indeed, the positive values suggest that an increase in the corresponding variable implies an increase in probability of being a fare evader and, *vice versa*, negative signs indicate the opposite condition. Nevertheless, before commenting on the extent of influence of the coefficients, a light modification to the model is introduced in order to present all variables in the categorical form (i.e. dichotomous and polychotomous). Indeed, *Month_freq* is a continuous variable. Therefore, for reasons of consistency, we want to verify whether by replacing *Month_freq* with a dichotomous variable, related to the daily frequency of

travel, the predictive ability of the model is retained or improved. Then, we introduce the variable *Daily_trips_over2*, which takes value 1 if the user makes more than two trips per day by bus, otherwise its value is 0 (dummy variable) in the opposite condition.

3.2.3. The refined model

Table 3 shows that the predictive ability of the model remains about the same (the *d.r.* decreases to a negligible extent from 47.83 to 47.56), as does the significance of the coefficients. Therefore, in order to comment on the influence on the propensity to evade of each independent variable, we will refer to this model and its results in deriving the fare-evader "profile", according to socio-demographic determinants, travel behaviour characteristics and personal knowledge of fare evasion.

3.2.3.1. The effect of socio-demographic determinants. The possible dodger is male. Indeed, a woman (*Gen_F*) is 1.48 times (i.e. 1/0.6758) less likely to be a fare evader. This result confirms previous Italian studies on fare evasion, which showed that the fare evaders are – largely – male (Abrate et al., 2008; Bucciol et al., 2013). Moreover, this outcome is consistent with the work of Eddy (2010), even if he concentrates his efforts on transit systems with barriers. In addition, this result confirms a previous experimental general study on deception, where it was shown that men are more likely than women to engage in deceptive behaviour, in order to secure a monetary benefit (Dreber and Johannesson, 2008). However, it is worth pointing out that, in the CTM bus service, the percentage of female users exceeds that of males (about 60% female compared to 40% male).

The possible dodger is young, below 26 years old. Being over 26 decreases the propensity to evade to different extents. Indeed, the age group 26–50 corresponds to an OR of 0.3292, whose inverse is equal to 3.038. That is, being between 26 and 50 decreases the probability of being a dodger 3.038-fold, than being under 26 and being over 50 reduced the probability of being a fare evader to 1/0.1127 = 8.87 times. This result is in accordance with previous Italian studies. In fact, both Abrate et al. (2008) and Bucciol et al. (2013) showed that holding no valid ticket is more likely in younger than older people. Conversely, Eddy (2010) reveals different findings compared to the canonical patterns of normal human behaviour. In particular, he shows that 68% of the people observed committing fare evasion were adults and only 22% were youths. A possible explanation for younger people to the detriment older people is the perception of a larger social "stigma" from older passengers, related to the possibility of being caught without a ticket.

The possible dodger has a low level of education. For example, a passenger with a middle school diploma -*Middle_sc*- has a probability to evade of 2.46 (=1/0.4061) times lower than those who did not reach this level of education. In addition, getting an upper school diploma -*Upper_sc*- has an effect on the probability to evade even lower than 3.81. This outcome differs to the one of Bucciol et al. (2013), where no effect of education level was found. This can be explained considering that a person with a high level of education perceives a certain social "stigma": it is probable that a highly-educated person is aware that to evade the fare prejudices other people.

The possible dodger is unemployed and/or a student. In fact, the *Unempl_y* and *Stud_y* determinants increase the probability of evading the fare by 1.485 and 1.415 times, respectively, compared to not being either unemployed or a student. These outcomes confirm previous Italian studies. Abrate et al. (2008) showed that housewives and the retired are less prone than students to evasion, while, on the other hand, the condition of being unemployed is associated with a larger level of evasion compared with students (+8%). Bucciol et al. (2013) showed that holding no valid ticket is more likely if the passenger is unemployed (30.5%). All these outcomes can be justified due to the economic crisis affecting Italy in recent years: more workplaces have been lost (CNEL, 2013).

The possible dodger travels by bus as no alternatives to the bus are feasible. Indeed, if a passenger can choose among different travels options (e.g. by car) his probability of evading the fare is lower than 1.61 times (=1/0.6211) compared to a passenger who travels by bus, because it represents his only travel option. That is, the possible fare evader has no trip options and travels by bus not by choice, but as he cannot do otherwise. This outcome partially contrasts the finding of Bucciol et al. (2013). In fact, even if the attribute is significant, they found that the probability of evading the fare works in an opposite way: having no travel options to the bus (and, therefore, being forced to use it) decreases the likelihood of evading. However, our findings can be supported considering that those who have multiple options in the context investigated, probably use the bus in some contingency situations (i.e. car breakdown), and so could accept paying the trip fare. In other words, the bus is not the first option, thus as passengers they could not know the strengths and weaknesses, and so would travel more relaxed by paying the fare.

3.2.3.2. The effect of travel behaviour determinants. Looking at travel behaviour determinants, only three variables, discussed below, are sufficient to explain the propensity to evade the fare.

The possible dodger makes short trips. Indeed, *In_vehicle_time_more15* affects the propensity to evade and, as one would expect, passengers who travel longer than 15 min have a 1.41 times (=1/0.7065) lower probability to evade the fare compared to those who undertake shorter trips. This finding is in accordance with Bucciol et al. (2013), even if, their study considered 30 min as a threshold to distinguish a short trip from a long one. Travelling for a short time seems a relevant predictor of the probability of holding no ticket (*p-value* < 0.001). Nevertheless, in short trips, our result can be justified considering that the possibility of having the ticket checked by the inspectors is reduced. Moreover, this result confirms a previous theoretical consideration discussed in Barabino et al. (2014).

The possible dodger makes more than two trips in the day, revealing that the dodger type is more likely to be a frequent passenger, rather than an occasional one. The influence of being a fare evader is positive. Indeed, a passenger who undertakes more than two trips per day by bus, evades 1.251 times more than those who travel less than twice per day. This result contrasts with that of Bucciol et al. (2013), who found that travel frequency is not a significant predictor in their reduced model (including fewer variables), while it becomes quite significant (p-value < 0.05) in their full model. Moreover, being occasional travellers increases the probability of evasion, contrary to our findings. However, our result can be supported by the fact that those who frequently travel by bus know the strengths and weaknesses. Therefore, as an example, one can know that the inspection level is modest, as is the probability of being caught evading the fare. Thus, being a systematic traveller supports our results.

The possible dodger is not satisfied with the service. This is an expected result. Indeed, a passenger satisfied with the transport service is 3.36 (=1/0.2979) times more likely not to be an evader compared to a dissatisfied one. By contrast, in Bucciol et al. (2013), being satisfied with the service or not is not a very significant (p-value <0.1) predictive variable. Moreover, *Satisf_y* affects the probability of evading the fare in an opposite way compared to our observations (i.e. being dissatisfied with the service reduces the probability of evasion).

3.2.3.3. The effect of personal knowledge of fare evasion. Focusing on the individual propensity to evade the fare, one observes that almost all the determinants have a significant effect in explaining the probability of being a fare evader. Moreover, in discussing the result, these considerations arise.

The possible dodger pays attention to the perceived frequency of inspection in a proportional way, at least in our context. This is an expected result. The higher is the perceived inspection frequency, the lower the propensity to evade. Indeed, a passenger who sees the inspectors engaged in inspection of tickets – *Insp_freq_1_5* – (i.e. from 1 and 5 times – in 4 months) has a probability to evade the fare 1.27 times lower than those who have never seen the inspectors on board, in the same time window. If the perceived frequency is higher (i.e. larger than five times), the passenger will tend to evade 1.56 times (=1/0.6413) less than those who have never seen the inspectors in the same time window. In other words, we can conclude that a passenger evades more easily if he has a low perception of the control of the ticket. This result is in accordance with the literature on deterrence (e.g. Beyleveld, 1980; Clarke et al., 2010; Von Hirsch et al., 1999), where it has arisen that potential offenders pay greater attention to the certainty of being caught than to the severity of the punishment if caught. Although our obtained outcomes were expected, they contrast with the ones of Bucciol et al. (2013), even if they use a slightly different variable.

The possible dodger knows the amount of the fine. Indeed, as shown by the determinant *Know_fine_y*, if a passenger knows the extent of the disbursement if caught evading the fare, he is 2.698 times more likely to be an evader. In our case study, even if somewhat strange, when a passenger is caught evading the fare it can depend on favourable situations. Indeed, the amount of the fine is roughly 60 times the fare, but if the fine is paid within 60 days of notification, it is reduced by two third, resulting in 20 times the evaded fare. Therefore, in our context, when both the level of inspection and the fine are low, the percentage of passengers inclined to evade is probably high (Barabino et al., 2013a, 2014). Although Bucciol et al. (2013) addressed the issue of the amount of the fine from a different perspective and showed that the amount of the fine is a not a very predictive variable (p-value <0.1), our outcomes can be considered in line with the findings of their model (in terms of coefficient signs), but, differently, the knowledge of the fine greatly affects the probability of being a possible evader.

The possible dodger will evade if he has already been fined in the past. In this case, the *Fine_past_y* represents the determinant which greatly affects the propensity of being an evader. The positive coefficient indicates that having already been fined in the past increases the likelihood of evading the fare, for a passenger. Moreover, reading the OR, we can say that having already been fined in the past increases the likelihood of evasion, 2.856 times, compared with never having been fined. This partially contrasts with the results of Bucciol et al. (2013), who showed that having already been fined in the past was not a significant predictor variable, even if the coefficient works as in our model. These different outcomes suggest that in the metropolitan area of Cagliari, fare evaders seem to perceive a context where fare evasion is not sufficiently strengthened by administrative issues. As a result, even if caught, the evader could leave the fine unpaid, as he will not be prosecuted for his violation after a fixed time window. Moreover, this outcome suggests that some probable fare evaders are chronic evaders, that is, they are evaders irrespective of the actions to curb fare evasion. Nevertheless, looking jointly at these results, the PTC can concentrate their efforts to curb fare evasion by strengthening inspections rather than working on the value of the fine.

To conclude, even though these new outcomes allow a more refined portrait of a possible fare evader to be drawn, outcomes of Bucciol et al. (2013) and Abrate et al. (2008) can be broadened, they can be considered preliminary and they partially remove the need for further investigations, which will be welcome when new data or new studies are available.

4. Conclusions and research perspectives

Although in *proof-of-payment* transit systems, fare evasion is a relevant concern that heavily affects PTCs, mainly in terms of lost fare revenues, damage to corporate image and reduced security, to the best of our knowledge, there are only two studies which have been undertaken to analyse who are fare evaders. These studies were undertaken in an Italian context, where the problem is particularly widespread. They evaluated the probability of being a fare evader using data directly derived from the ownership or not of the ticket. Therefore, in order to shed additional light on the characteristics of fare evaders

in a scenario, namely without ticket inspection activities, in a whole study we examined the influence of a set of independent variables, which affect the probability of being a fare evader. To derive the "profile" of the possible dodger, we grouped determinants into three classes: socio-demographic, travel behaviour and personal attitude to fare evasion. With about 2200 on-board personal interviews collected among passengers of an Italian PTC, we showed the influence of all determinants in evaluating the propensity of being a fare evader by LRMs. Although the research was small in scale (i.e. the dataset is not very large and includes data on the passengers of an Italian PTC) and further studies are recommended to confirm the results, it was large enough to establish patterns in the dodger "profile". Nevertheless, considering the socio-demographic determinants, we have shown that males, younger than 26 years, with low levels of education, unemployed and/or students and without alternatives to using the bus are more probably free riders. Moreover, looking at travel behaviour determinants, people who make trips of less than 15 min, who are systematic passengers and are not satisfied with the service, are possible fare evaders. In addition, considering the individual propensity to evade the fare, we found that a low level of inspection, the knowledge of the fine and previous ticket violations are determinants which make people more probable to evade the fare.

To the best of our knowledge, this is the first empirical study which (i) faced fare evasion by evaluating the propensity to be a fare evader in probabilistic terms, and (ii) considered more variables than previous works. Moreover, even if this study draws a portrait of the fare evader in the context of Cagliari, it contributes to a clearer sketch of the determinants which affect the probability of being a fare evader. It also shows that for PTCs it seems better to focus on the inspection level, rather than on the amount of the fine, as a possible option for curbing fare evasion. In addition, these results can be used as a starting point to implement strategies addressing these subjects and to evaluate the effect they produce. For example, the PTC can consider some actions such as using moral reminders, and optimizing the inspection with a sub-set of passengers. Next, PCTs can verify the effects of these actions on the propensity of being a possible fare evader.

Nevertheless, this is an initial step in the research. In the electronic age it would be interesting to evaluate the effects on the propensity of fare evasion by introducing near field communication cards and forcing all passengers to validate their tickets when boarding. This *modus operandi* is expected to reduce the fare evasion rate and thus the probability of evading the fare. Indeed, a "social check" function on dodgers could be activated: non-evader passengers could label evader passengers as cheats. As a result, a mechanism based on social "stigma" is expected to be generated.

Acknowledgments

The authors are also grateful to the General Management of CTM SpA for its support in this research and the possibility to use its real data for the experimentation. The views expressed in this paper are those of the authors only. The authors are also grateful to anonymous reviewer, who helped write a better paper.

References

Abeler, J., Becker, A., Falk, A., 2014. Representative evidence on lying costs. J. Public Econom. 113, 96–104.
Abrate, G., Fraquelli, G., Meko, E., Rodia, G., 2008. L'Evasione Tariffaria nel Trasporto Pubblico Locale: un'Analisi Empirica. Conferenza Società Italiana di Economia Pubblica, XX Riunione Scientifica. Pavia, 25–26 Settembre 2008.
Avenhaus, R., 2004. Applications of inspection games. Math. Modell. Anal. 9 (3), 179–192.
Barabino, B., Salis, S., Useli, B., 2013a. A modified model to curb fare evasion and enforce compliance: empirical evidence and implications. Transp. Res. Part A 58, 29–39.
Barabino, B., Deiana, E., Tilocca, P., 2013b. Face to face and telephone interviews as competing yet complementary methods to elicit information. In: Proceedings of Urban Transport XIX, May 2013, Kos, Greece, pp. 647–663. doi: http://dx.doi.org/10.2495/UT130521.
Barabino, B., Salis, S., Useli, B., 2014. Fare evasion in *proof-of-payment* transit systems: deriving the optimum inspection level. Transp. Res. Part B 70, 1–17.
Beyleveld, D.A., 1980. A Bibliography on General Deterrence Research. Saxon House, Westmead, UK.
Bijleveld, C., 2007. Fare dodging and the strong arm of the law. J. Exp. Criminology 3 (2), 183–199.
Bonfanti, G., Wagenknecht, T., 2010. Human factors reduce aggression and fare evasion. Public Transp. Int. 59 (1), 28–32.
Bootheway, G.B.P., 2009. On the optimality of fines when enforcement is risky. ASBBS E – J. 5 (1), 33–39.
Borndörfer, R., Buwaya, J., Sagnol, G., Swarat, E., 2013. Optimizing toll enforcement in transportation networks: a game-theoretic approach. Electron. Notes Discrete Math. 41, 253–260.
Borndörfer, R., Omont, B., Sagnol, G., Swarat, E., 2012. A Stackelberg game to optimize the distribution of controls in transportation networks. In: Game Theory for Networks. Springer, Berlin Heidelberg, pp. 224–235.
Boyd, C., Martini, C., Rickard, J., Russell, A., 1989. Fare evasion and non-compliance: a simple model. J. Transp. Econom. Policy 23 (2), 189–197.
Bucciol, A., Landini, F., Piovesan, M., 2013. Unethical behaviour in the field: demographic characteristics and beliefs of the cheater. J. Econ. Behav. Organ. 93, 248–257.
Clarke, R.V., Contre, S., Petrossian, G., 2010. Deterrence and fare evasion: results of a natural experiment. Security J. 23 (1), 5–17.
CNEL – Consiglio Nazionale Economia del Lavoro, 2013. Rapporto sul mercato del lavoro 2012–2013. <http://www.cliclavoro.gov.it/Barometro-Del-Lavoro/Documents/CNEL_Rapporto_sul_mercato_del_lavoro_2012-2013.pdf>.
Correa, J.R., Harks, T., Kreuzen, V.J., Matuschke, J., 2014. Fare Evasion in Transit Networks. Preprint, arXiv:1405.2826.
Denstadli, J.M., 2000. Analyzing air travel: a comparison of different survey methods and data collection procedures. J. Travel Res. 39 (1), 4–10.
Dreber, A., Johannesson, M., 2008. Gender difference in deception. Econom. Lett. 99 (1), 197–199.
Eddy, D., 2010. Fare evasion – is it a youth issue? Transit Australia – Australia's Urban Passenger Transp. J. 65 (12), 1–7.
Erard, B., Feinstein, J.S., 1994. Honesty and evasion in the tax compliance game. Rand J. Econ., 1–19
Gneezy, U., 2005. Deception: the role of consequences. Am. Econ. Rev. 95 (1), 384–394.
Killias, M., Scheidegger, D., Nordenson, P., 2009. The effects of increasing the certainty of punishment: a field experiment on public transportation. Eur. J. Criminology 6 (5), 387–400.
Kooreman, P., 1993. Fare evasion as a result of expected utility maximisation. Some empirical support. J. Transp. Econom. Policy 27 (1), 69–74.
Maskin, E.S., 1977. Nash Equilibrium and Welfare Optimality. MIT Discussion Paper.
Matsushima, H., 2008. Role of honesty in full implementation. J. Econom. Theory 139 (1), 353–359.
Mazar, N., Amir, O., Ariely, D., 2008. The dishonesty of honest people: a theory of self-concept maintenance. J. Mark. Res. 45 (6), 633–644.

Mehlkop, G., Becker, R., Imhof, R., 2007. Extending the rational action theory of crime: success, norms and risk-taking. An empirical re-assessment of the theory of subjective expected utility. Proceedings of the International Conference on Rational Choice and Social Institutions, Zurich, 6–8 September.
Multisystems Inc., Mundle & Associates Inc., Parsons Transportation Group Inc., 2002. A Toolkit for Self-Service, Barrier-Free Fare Collection. Transit Cooperative Research Program, Report 80. TRB, Washington D.C.
Porst, R., Von Briel, C., 1995. Wären Sie vielleicht bereit, sich gegebenenfalls noch einmal befragen zu lassen?: oder: Gründe für die Teilnahme an Panelbefragungen.
Reddy, A.V., Kuhls, J., Lu, A., 2011. Measuring and controlling subway fare evasion. Transp. Res. Rec. 2216, 85–99.
Sasaki, Y., 2014. Optimal choices of fare collection systems for public transportations: barrier versus barrier-free. Transp. Res. Part B 60, 107–114.
Simmons, E., Wilmot, A., 2004. Incentive payments on social surveys: a literature review. Social Survey Meth. Bull., 1–11
Singer, E., Bossarte, R.M., 2006. Incentives for survey participation: When are they "coercive"? Am. J. Prev. Med. 31 (5), 411–418.
Singer, E., Van Hoewyk, J., Gebler, N., Raghunathan, T., McGonagle, K., 1999. The effects of incentives on response rates in interviewer-mediated surveys. J. Official Statist. 15 (2), 217–230.
Singer, E., Ye, C., 2013. The use and effects of incentives in surveys. ANNALS Am. Acad. Political Social Sci. 645 (1), 112–141.
Suquet, J.B., 2010. Drawing the line: how inspectors enact deviant behaviors. J. Service Marketing 24 (6), 468–475.
Richardson, A.J., Ampt, E.S., Meyburg, A.H., 1995. Survey Methods for Transport Planning. Eucalyptus Press, Melbourne, p. 314.
Transit Cooperative Research Program (TCRP), Synthesis 63, 2005. On Board and Intercept Transit Survey Techniques – A Synthesis of Transit Practice, Transportation Research Board, Washington, DC.
Traxler, C., 2010. Social norms and conditional cooperative taxpayers. Eur. J. Polit. Econ. 26 (1), 89–103.
Von Hirsch, A., Bottoms, A.E., Burney, E., Wikström, P.-O.H., 1999. Criminal Deterrence and Sentence Severity: An Analysis of Recent Research. Hart Publishing, Oxford, UK.
Yin, Z., Jiang, A.X., Johnson, M., Tambe, M., Kiekintveld, C., Leyton-Brown, K., Sandholm, T., Sullivan, J., 2012. TRUSTS: scheduling randomized patrols for fare inspection in transit systems using game theory. AI Mag. 33 (4), 59–72.